Case 02 · Ideation document: Fire-Sale Cascade Simulator

Following chapter 1 (Initiation) of the Magshimim project-book template. For each project: overview, goal, target audience, existing solutions, technological core, technologies, challenges and sources, hardware requirements, and conclusion.

Overview

A tool that models an entire financial market as a set of "agents" (pension funds, banks, credit funds, each with its own balance sheet and behavioural rules) using agent-based modelling, in order to see what happens when a rate rise is injected: who is forced to sell, and how their selling pushes the next one in line.

In practice: the user builds or loads a "world" (say Britain in September 2022), moves the rate slider, and watches step by step: bond prices fall, a leveraged fund gets a margin call, sells bonds to meet it, the selling pushes prices lower still, and now the next fund gets a call too. The tool marks the point of no return, and the place where an intervention (such as the central bank's) breaks the loop.

Project goal

We chose this project because it is the most important thing in the story, and also the hardest to see: the loop that feeds itself. The story mentions two cases (British pension funds and SVB), and in both it was not the rate rise alone that brought the system down, but the chain reaction it set off. People understand "rates went up". They find it hard to picture "one sale caused three more".

The goal: a tool in which a past crisis can be reproduced, the model checked against reality, and then "what if" questions asked about today's world: for example, what happens if leveraged private credit behaves like the LDI funds.

Target audience

- **Primary audience:** financial-stability analysts (at central banks, supervisors and risk departments), with good knowledge of balance sheets and leverage. They need a quick "what if" tool before running a heavy official model.
- **Secondary audience:** lecturers and students of financial economics who need to show a leveraged unwind in a way that can be seen. And journalists who want to explain a crisis.
- **Illustrative scenario:** an analyst at a supervisor reads that private-credit defaults are rising. He builds a world with 40 leveraged credit funds, injects a 1% rate rise, and checks how many of them reach a forced-sale point, and in what order.

Existing solutions

- **Bank of England SWES (System-Wide Exploratory Scenario, 2023 to 2024):** the first official exercise of its kind, testing how banks, funds and insurers respond together to a shock. Closest in spirit, but a one-off exercise with confidential data, not a tool.

- **IMF (FSAP) and central-bank stress tests:** test each institution alone against a scenario, and much less the interaction between them. That is exactly the weakness crises expose.
- **Academic research:** the Greenwood, Landier and Thesmar model ("Vulnerable Banks", 2015) of fire sales between banks, and work by Cont and Schaanning on common price impact. They supply the mathematics, not a tool to play with.
- **Mesa (a Python ABM library):** general infrastructure for agent simulations. A building block, not a solution.

What sets us apart: an open tool that connects the academic models to an interactive "what if" interface, with calibrated reproductions of real crises as a test bench.

Technological core

The core is the simulation engine: a loop that at each time step (1) updates asset prices according to excess selling (price impact), (2) recomputes every agent's balance sheet and leverage, (3) fires the response rules (collateral calls, sales, halted lending), and (4) feeds the new sales into the next step. The key mechanism is the price-impact function: how far a bond's price falls when a given quantity is sold at once.

Without this engine there is no project: reproductions, scenarios and visualisation are its inputs and outputs.

Main technologies

- **Python with NumPy:** all agents' balance sheets are held as arrays, so a simulation step is a few matrix operations rather than a loop over agents. That is what allows thousands of agents and thousands of runs.
- **NetworkX:** for the exposure network (who shares assets with whom, who owes whom). A network lets us identify "central nodes" whose fall brings down the most.
- **Pyodide:** so the simulation runs in the browser with no server. This is also what worked in the Case 01 POC.
- **Canvas or D3.js:** to animate the loop step by step.

Open questions:

- **Mesa versus a vectorised engine of our own:** Mesa gives a ready, convenient structure but works agent by agent and is slow at scale. A vectorised engine is faster but takes more work. We will decide after measuring in the POC.
- **How complex should behavioural rules be:** simple rules (sell when leverage crosses a threshold) are transparent and easy to test. Complex rules are more realistic, but it becomes hard to know why the model did what it did.

Technological challenges and sources

- **Calibration:** how do we set the parameters (leverage, collateral thresholds, price impact) so the model reproduces Britain 2022? We will use the data the Bank of England published on the LDI crisis and on SWES, and check that the model reaches the same direction and order of magnitude, not the exact same number.

- **Missing exposure data:** who holds how much is precisely the hidden part. We will build the tool to work with distributions ("between 20% and 40% of funds are leveraged above 3x") and show the range of outcomes, rather than pretend to have an exact number.
- **Sensitivity:** such models can be very sensitive to a single parameter. We will build in a sensitivity check that shows which parameter moves the result the most.
- **Browser performance:** thousands of runs in Pyodide may be slow. We will measure in the POC.

Sources: Mesa documentation (mesa.readthedocs.io), the Bank of England's financial-stability reports on LDI and SWES, and the paper by Greenwood, Landier and Thesmar.

Hardware requirements

None. The simulation runs in a browser or on an ordinary computer. Heavy calibration runs can use a stronger machine, but no special hardware is needed.

Conclusion

✅ Survives, with a condition. The idea goes straight to the heart of the story, and it has an academic anchor and two real crises to test against. The condition: data. In the research stage we must show that even with distributions instead of exact exposures, the tool gives a useful answer and not just "anything is possible".